

Q2,

noting that ref. frame and res. frame are the same:

$$\underline{r}_{eb}^e = \underline{r}_{ib}^e - \underline{r}_{ie}^e$$

$$\dot{\underline{r}}_{eb}^e = \dot{\underline{r}}_{ib}^e = \underline{\alpha}_{eb}^e$$

To find $\dot{\underline{r}}_{eb}^e$, we first need to define $\underline{r}_{eb}^e$.

$$= \underline{C}_{ib}^e \underline{r}_{ib}^i$$

Differentiating:

$$\Rightarrow \underline{r}_{eb}^e = \underline{C}_{ib}^e \underline{r}_{ib}^i + \underline{C}_{ib}^e \dot{\underline{r}}_{ib}^i$$

$$= -\Omega_{ie}^e \underline{C}_{ib}^e \underline{r}_{ib}^i + \underline{C}_{ib}^e \dot{\underline{r}}_{ib}^i$$

Differentiating again

$$\Rightarrow \ddot{\underline{r}}_{eb}^e = -\dot{\Omega}_{ie}^e \underline{C}_{ib}^e \underline{r}_{ib}^i - \Omega_{ie}^e \dot{\underline{C}}_{ib}^e \underline{r}_{ib}^i - \Omega_{ie}^e \underline{C}_{ib}^e \dot{\underline{r}}_{ib}^i + \underline{C}_{ib}^e \ddot{\underline{r}}_{ib}^i + \underline{C}_{ib}^e \dot{\underline{r}}_{ib}^i$$

$$\ddot{\underline{r}}_{eb}^e = 0 - \Omega_{ie}^e \Omega_{ie}^e \underline{C}_{ib}^e \underline{r}_{ib}^i - 2\Omega_{ie}^e \underline{C}_{ib}^e \dot{\underline{r}}_{ib}^i + \underline{a}_{ib}^e$$

earth's rotation is const., i.e. $\dot{\Omega}_{ie}^e = 0$

Now, recall the ECEF NE: ..., substituting for $\underline{a}_{ib}^e$, we get:

$$\Rightarrow \underline{v}_{eb}^e = \underline{f}_{ib}^e + \underline{g}_{ib}^e(r_{ib}^e) - \Omega_{ie}^e \Omega_{ie}^e \underline{r}_{ib}^e - 2\Omega_{ie}^e \underline{v}_{ib}^e$$

$$= \underline{C}_{ib}^e \underline{f}_{ib}^b + \underline{g}_{ib}^e(L, h) - 2\Omega_{ie}^e \underline{v}_{ib}^e$$

why $\underline{v}_{eb}^e = \underline{v}_{ib}^e$? proof: $\underline{v}_{eb}^e = \underline{v}_{ib}^e - \underline{v}_{ie}^e = \underline{v}_{ib}^e$

Q3,

$$\underline{v}_{eb}^n = \underline{C}_{en}^n \underline{v}_{eb}^e$$

differentiating: $\dot{\underline{v}}_{eb}^n = \dot{\underline{C}}_{en}^n \underline{v}_{eb}^e + \underline{C}_{en}^n \dot{\underline{v}}_{eb}^e$

applying the strapdown eq. and sub. for $\dot{\underline{v}}_{eb}^e$

$$= -\Omega_{en}^n \underline{C}_{en}^n \underline{v}_{eb}^e + \underline{C}_{en}^n (\underline{C}_{ib}^e \underline{f}_{ib}^b + \underline{g}_{ib}^e(L, h) - 2\Omega_{ie}^e \underline{v}_{ib}^e)$$

$$= \underline{C}_{ib}^n \underline{f}_{ib}^b + \underline{g}_{ib}^n(L, h) - 2\Omega_{ie}^e \underline{v}_{eb}^n - \Omega_{en}^n \underline{v}_{eb}^n$$

From $\underline{C}_{en}^e \Omega_{ie}^e \underline{C}_{in}^e = \Omega_{ie}^n \Rightarrow \underline{C}_{en}^e \Omega_{ie}^e = \Omega_{ie}^n \underline{C}_{in}^{e-1} = \Omega_{ie}^n \underline{C}_{en}^n$

therefore: $\dot{\underline{v}}_{eb}^n = \underline{C}_{ib}^n \underline{f}_{ib}^b + \underline{g}_{ib}^n(L, h) - (\Omega_{en}^n + 2\Omega_{ie}^n) \underline{v}_{eb}^n$

However, from $\dot{\underline{C}}_{ib}^n = \underline{C}_{ib}^n \Omega_{nb}^b$, $\underline{\omega}_{nb}^b$ is now measured

what we can measure is $\underline{\omega}_{ib}^b$, then

$$\underline{\omega}_{ib}^b = \underline{\omega}_{ie}^b + \underline{\omega}_{en}^b + \underline{\omega}_{nb}^b$$

$$\Rightarrow \underline{\omega}_{nb}^b = \underline{\omega}_{ib}^b - \underline{\omega}_{ie}^b - \underline{\omega}_{en}^b$$

$$C_b^n \Omega_{ie}^b C_n^b = \Omega_{ie}^n$$

$$\Rightarrow \dot{C}_b^n = C_b^n (\Omega_{ib}^b - \Omega_{ie}^b - \Omega_{en}^b)$$

$$\Rightarrow C_b^n \Omega_{ie}^b = \Omega_{ie}^n C_b^n$$

$$= C_b^n \Omega_{ib}^b - C_b^n (\Omega_{ie}^b + \Omega_{en}^b)$$

$$C_b^n \Omega_{en}^b C_n^b = \Omega_{en}^n$$

$$= C_b^n \Omega_{ib}^b - \Omega_{ie}^n C_b^n - \Omega_{en}^n C_b^n$$

$$= C_b^n \Omega_{ib}^b - (\Omega_{ie}^n + \Omega_{en}^n) C_b^n$$